

Отчёт по лабораторной работе № 7

Настройка межсетевого экрана

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2026-09-14

Содержание

1	Цель работы	5
2	Задание	6
3	Теоретические сведения	7
4	Выполнение работы	8
4.1	Пользовательский сервис ssh-custom	8
4.2	Разрешённые сетевые сервисы	9
4.3	Перенаправление порта	10
4.4	Маршрутизация IPv4	11
4.5	Маскарадинг и доступ клиента	12
4.6	Финальная проверка firewalld	13
5	Выводы	15

Список иллюстраций

4.1	Пользовательский сервис ssh-custom	9
4.2	Разрешённые сетевые сервисы	10
4.3	Перенаправление порта	11
4.4	Маршрутизация IPv4	12
4.5	Маскарадинг и доступ клиента	13
4.6	Финальная проверка firewalld	14

Список таблиц

1 Цель работы

Получить навыки настройки firewalld, перенаправления портов и маршрутизации.

2 Задание

1. Создано описание сервиса ssh-custom для TCP-порта 2022.
2. Сервисы HTTP, HTTPS и ssh-custom добавлены в постоянную конфигурацию.
3. Настроено перенаправление 2022/tcp на 22/tcp.
4. Включена IPv4-маршрутизация и маскердинг для внутренней сети.
5. Клиент успешно подключился к SSH через порт 2022 и получил доступ во внешнюю сеть.
6. Сценарий server-firewall закрепил настройки.

3 Теоретические сведения

`firewalld` управляет правилами фильтрации по зонам и сервисам. Маскарадинг подменяет исходный адрес пакетов, а `ip_forward` разрешает ядру маршрутизацию между интерфейсами.

4 Выполнение работы

4.1 Пользовательский сервис ssh-custom

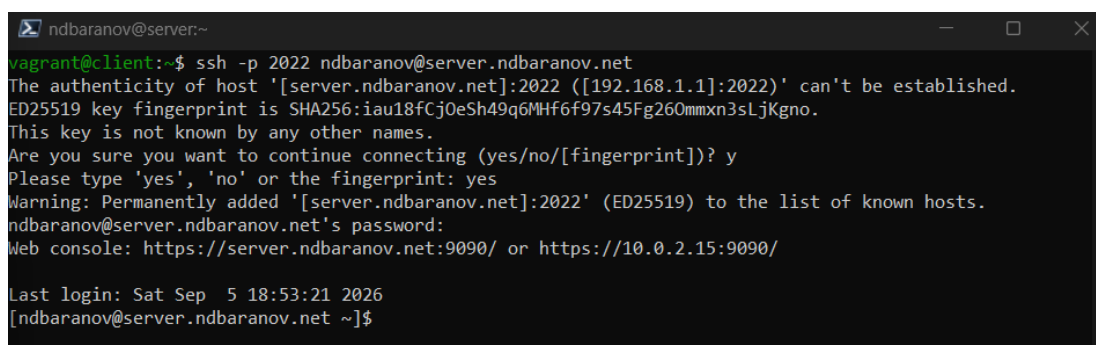
Создано описание firewalld-сервиса ssh-custom для TCP-порта 222. XML-файл позволяет управлять правилом по понятному имени.

```
root@server:/etc/firewalld/services
root@server:~# cp /usr/lib/firewalld/services/ssh.xml /etc/firewalld/services/ssh-custom.xml
cd /etc/firewalld/services/
root@server:/etc/firewalld/services# cat /etc/firewalld/services/ssh-custom.xml
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>SSH</short>
  <description>Secure Shell (SSH) is a protocol for logging into and executing commands on remote machines. It provides secure encrypted communications. If you plan on accessing your machine remotely via SSH over a firewalled interface, enable this option. You need the openssh-server package installed for this option to be useful.</description>
  <port protocol="tcp" port="22"/>
</service>
root@server:/etc/firewalld/services# nano /etc/firewalld/services/ssh-custom.xml
root@server:/etc/firewalld/services# nano /etc/firewalld/services/ssh-custom.xml
root@server:/etc/firewalld/services# nano /etc/firewalld/services/ssh-custom.xml
root@server:/etc/firewalld/services# root@server:/etc/firewalld/services#
root@server:/etc/firewalld/services# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp amqps anno-1602
anno-1800 apcupsd aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareo
s-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-export
er ceph-mon cfengine checkmk-agent civilization-iv civilization-v cockpit collectd condor-collector crat
edb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-quic dns-over-tls d
ocker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger forema
n foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gangli
a-client ganglia-master git gitea gpsd grafana gre high-availability http http3 https ident imap imaps i
perf2 iperf3 ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kib
ana klogin kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kub
e-controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler
-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-net
work llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcache minecraft minidlna mndp mon
godb mosh mountd mpd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbi
os-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storag
econsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus p
rometheus-node-exporter proxy-dhcp proxy-http ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radi
us radsec rdp redis redis-sentinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client
samba-dc sane settlers-history-collection sip sips slimevr slp smtp smtp-submission smtps snmp snmptls
snmptls-trap snmptrap socks spideroak-lansync spotify-sync squid ssdp ssh statsrv steam-lan-transfer ste
am-streaming stellaris stronghold-crusader stun stuns submission supertuxkart svdrp svn syncthing syncti
ing-gui syncthing-relay synergy syscomlan syslog syslog-ng syslog-tls telnet tentacle terraria tftp tile
38 tinc tor-socks transmission-client turn turns upnp-client vdsms vnc-server vrrp warpinator wbem-http w
bem-https wireguard ws-discovery ws-discovery-client ws-discovery-host ws-discovery-tcp ws-discovery-udp
wsdd wsdd-http wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-java
-gateway zabbix-server zabbix-trapper zabbix-web-service zero-k zerotier
root@server:/etc/firewalld/services# cat /etc/firewalld/services/ssh-custom.xml
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>SSH custom</short>
  <description>Secure Shell (SSH) is a protocol for logging into and executing commands on remote machines. It provides secure encrypted communications. If you plan on accessing your machine remotely via SSH over a firewalled interface, enable this option. You need the openssh-server package installed for this option to be useful.</description>
  <port protocol="tcp" port="2022"/>
</service>
root@server:/etc/firewalld/services#
```

Рисунок 4.1: Пользовательский сервис ssh-custom

4.2 Разрешённые сетевые сервисы

В постоянную конфигурацию зоны добавлены HTTP, HTTPS и ssh-custom. После reload они отображаются в списке services.



```
ndbaranov@server:~  
vagrant@client:~$ ssh -p 2022 ndbaranov@server.ndbaranov.net  
The authenticity of host '[server.ndbaranov.net]:2022 ([192.168.1.1]:2022)' can't be established.  
ED25519 key fingerprint is SHA256:iau18fCj0eSh49q6MHf6f97s45Fg260mmxn3sLjKgn0.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? y  
Please type 'yes', 'no' or the fingerprint: yes  
Warning: Permanently added '[server.ndbaranov.net]:2022' (ED25519) to the list of known hosts.  
ndbaranov@server.ndbaranov.net's password:  
Web console: https://server.ndbaranov.net:9090/ or https://10.0.2.15:9090/  
  
Last login: Sat Sep  5 18:53:21 2026  
[ndbaranov@server.ndbaranov.net ~]$
```

Рисунок 4.2: Разрешённые сетевые сервисы

4.3 Перенаправление порта

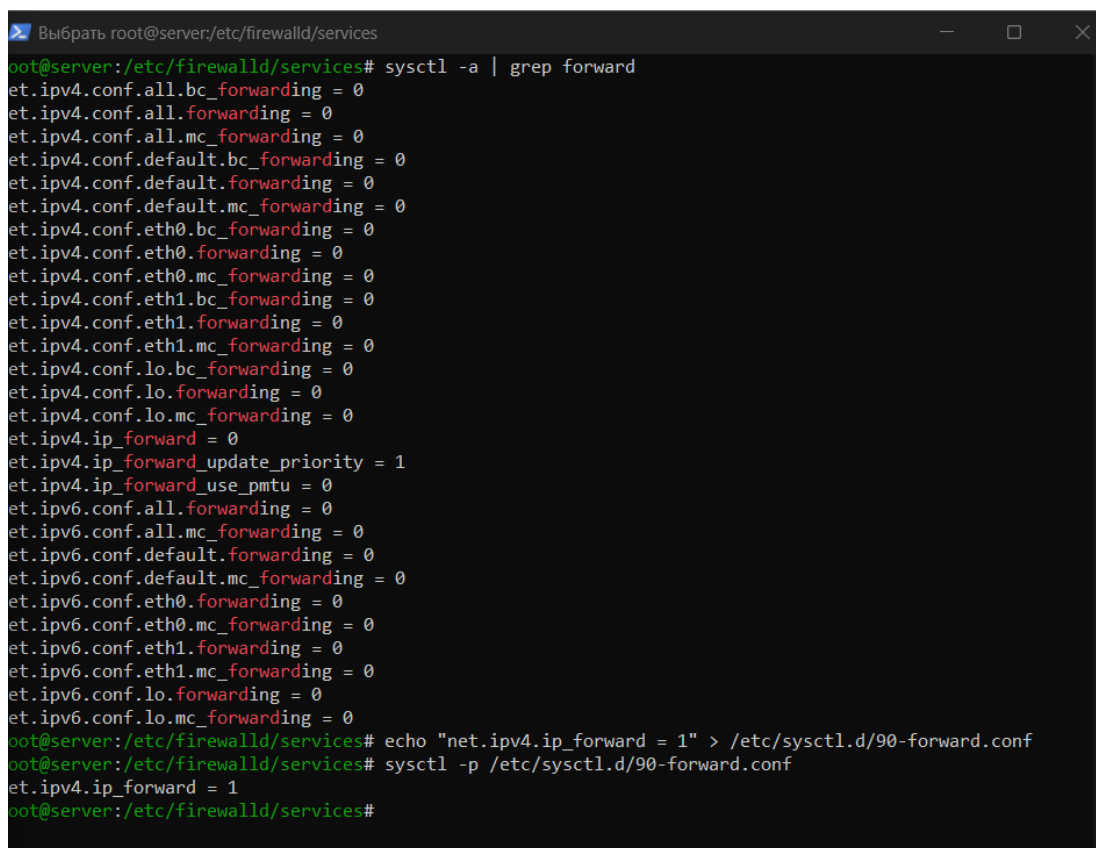
Настроено перенаправление входящих соединений с 2022/tcp на стандартный SSH-порт 22. Правило позволяет изменить внешний порт без перенастройки sshd.

```
Выбрать root@server:/etc/firewalld/services
root@server:/etc/firewalld/services# firewall-cmd --reload
success
root@server:/etc/firewalld/services# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp amqps anno-1602
anno-1800 apcupsd aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareo
s-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-export
er ceph-mon cfengine checkmk-agent civilization-iv civilization-v cockpit collectd condor-collector crat
edb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-quit dns-over-tls d
ocker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger forema
n foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gangli
a-client ganglia-master git gitea gpsd grafana gre high-availability http http3 https ident imap imaps i
perf2 iperf3 ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kib
ana klogin kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kub
e-controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler
-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-net
work llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcache minecraft minidlna mndp mon
godb mosh mountd mpd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbi
os-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storag
econsole ovirt-vmconsole plex pncd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus p
rometheus-node-exporter proxy-dhcp proxy-http ps2link ps3netshr ptp pulseaudio puppetmaster quassel radi
us radsec rdp redis redis-sentinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client
samba-dc sane settlers-history-collection sip sips slimevr slp smtp smtp-submission smtps snmp snmptls
snmptls-trap snmptrap socks spideroak-lansync spotify-sync squid ssdp ssh ssh-custom statsrv steam-lan-t
ransfer steam-streaming stellaris stronghold-crusader stun stuns submission supertuxkart svdrp svn synct
hing syncthing-gui syncthing-relay synergy syscomlan syslog syslog-ng syslog-tls telnet tentacle terrari
a tftp tile38 tinc tor-socks transmission-client turn turns upnp-client vdsd vnc-server vrrp warpinator
wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-host ws-discovery-tcp ws-di
scovery-udp wsdd wsdd-http wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent
zabbix-java-gateway zabbix-server zabbix-trapper zabbix-web-service zero-k zerotier
root@server:/etc/firewalld/services# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https ssh
root@server:/etc/firewalld/services# firewall-cmd --add-service=ssh-custom
firewall-cmd --list-services
success
cockpit dhcp dhcpv6-client dns http https ssh ssh-custom
root@server:/etc/firewalld/services# firewall-cmd --add-service=ssh-custom --permanent
firewall-cmd --reload
success
success
root@server:/etc/firewalld/services# firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22
success
root@server:/etc/firewalld/services#
```

Рисунок 4.3: Перенаправление порта

4.4 Маршрутизация IPv4

Параметр `net.ipv4.ip_forward` включён через `sysctl`. Сервер теперь может пересылать пакеты между своими интерфейсами.

A terminal window titled 'Выбрать root@server:/etc/firewalld/services' displays the output of the command 'sysctl -a | grep forward'. The output lists various sysctl parameters related to IPv4 and IPv6 forwarding, all set to 0. The user then enters 'echo "net.ipv4.ip_forward = 1" > /etc/sysctl.d/90-forward.conf', followed by 'sysctl -p /etc/sysctl.d/90-forward.conf', which updates 'net.ipv4.ip_forward' to 1.

```
root@server:/etc/firewalld/services# sysctl -a | grep forward
et.ipv4.conf.all.bc_forwarding = 0
et.ipv4.conf.all.forwarding = 0
et.ipv4.conf.all.mc_forwarding = 0
et.ipv4.conf.default.bc_forwarding = 0
et.ipv4.conf.default.forwarding = 0
et.ipv4.conf.default.mc_forwarding = 0
et.ipv4.conf.eth0.bc_forwarding = 0
et.ipv4.conf.eth0.forwarding = 0
et.ipv4.conf.eth0.mc_forwarding = 0
et.ipv4.conf.eth1.bc_forwarding = 0
et.ipv4.conf.eth1.forwarding = 0
et.ipv4.conf.eth1.mc_forwarding = 0
et.ipv4.conf.lo.bc_forwarding = 0
et.ipv4.conf.lo.forwarding = 0
et.ipv4.conf.lo.mc_forwarding = 0
et.ipv4.ip_forward = 0
et.ipv4.ip_forward_update_priority = 1
et.ipv4.ip_forward_use_pmtu = 0
et.ipv6.conf.all.forwarding = 0
et.ipv6.conf.all.mc_forwarding = 0
et.ipv6.conf.default.forwarding = 0
et.ipv6.conf.default.mc_forwarding = 0
et.ipv6.conf.eth0.forwarding = 0
et.ipv6.conf.eth0.mc_forwarding = 0
et.ipv6.conf.eth1.forwarding = 0
et.ipv6.conf.eth1.mc_forwarding = 0
et.ipv6.conf.lo.forwarding = 0
et.ipv6.conf.lo.mc_forwarding = 0
root@server:/etc/firewalld/services# echo "net.ipv4.ip_forward = 1" > /etc/sysctl.d/90-forward.conf
root@server:/etc/firewalld/services# sysctl -p /etc/sysctl.d/90-forward.conf
et.ipv4.ip_forward = 1
root@server:/etc/firewalld/services#
```

Рисунок 4.4: Маршрутизация IPv4

4.5 Маскарадинг и доступ клиента

В зоне включён masquerade, после чего клиент получил доступ через сервер. Проверка соединения показывает работу трансляции исходных адресов.

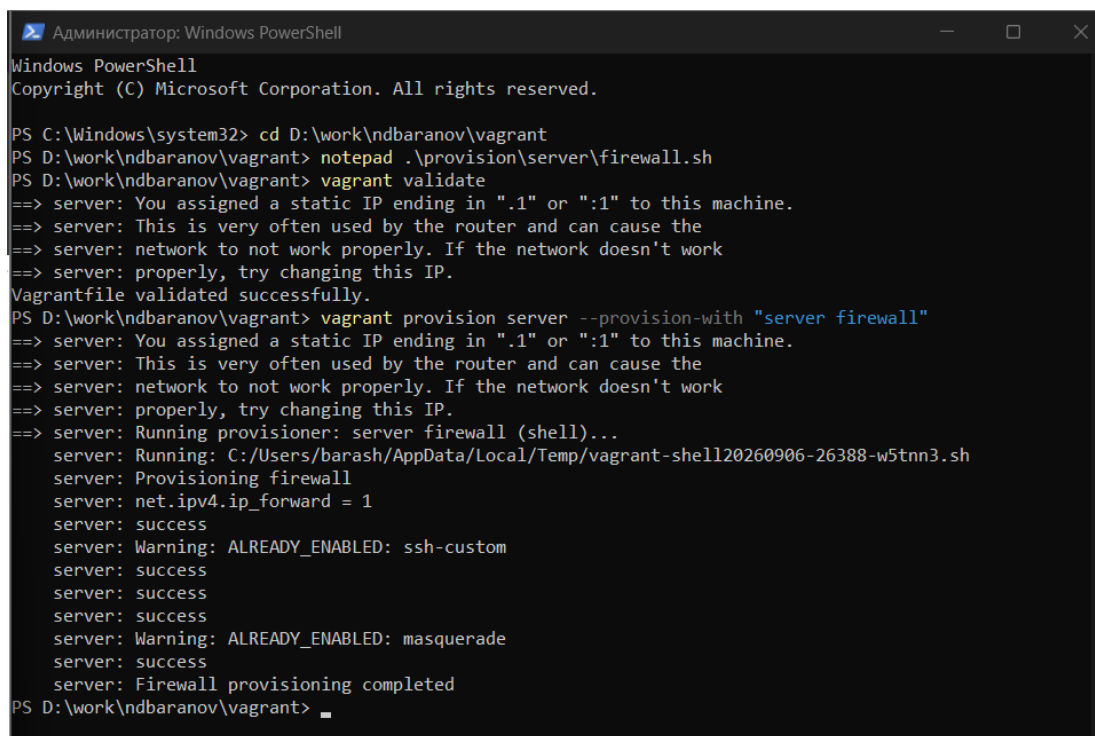
```
ndbaranov@server:~$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
default via 192.168.1.1 dev eth1 proto static metric 101
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100
192.168.1.0/24 dev eth1 proto kernel scope link src 192.168.1.1 metric 101
ndbaranov@server.ndbaranov.net ~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=18.3 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=19.1 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=17.7 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=17.6 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3025ms
rtt min/avg/max/mdev = 17.636/18.181/19.089/0.580 ms
ndbaranov@server.ndbaranov.net ~$
```

Рисунок 4.5: Маскарадинг и доступ клиента

4.6 Финальная проверка firewallld

Выведена итоговая конфигурация зоны и выполнен provision-сценарий. В результате правила сохраняются после перезагрузки и воспроизводятся автоматически.



```
Администратор: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Windows\system32> cd D:\work\ndbaranov\vagrant
PS D:\work\ndbaranov\vagrant> notepad .\provision\server\firewall.sh
PS D:\work\ndbaranov\vagrant> vagrant validate
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
Vagrantfile validated successfully.
PS D:\work\ndbaranov\vagrant> vagrant provision server --provision-with "server firewall"
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Running provisioner: server firewall (shell)...
server: Running: C:/Users/barash/AppData/Local/Temp/vagrant-shell20260906-26388-w5tnn3.sh
server: Provisioning firewall
server: net.ipv4.ip_forward = 1
server: success
server: Warning: ALREADY_ENABLED: ssh-custom
server: success
server: success
server: success
server: Warning: ALREADY_ENABLED: masquerade
server: success
server: Firewall provisioning completed
PS D:\work\ndbaranov\vagrant>
```

Рисунок 4.6: Финальная проверка firewalld

5 Выводы

firewalld фильтрует доступ к службам, перенаправляет порт 2022 на SSH и обеспечивает маршрутизацию внутренней сети через маскардинг.